Meningioma Growth Forecasting from Multi-Modal 3D MRI: A Three-Stage Complexity Ladder

From Volumetric Baselines to Foundation Model Features

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Abstract

Meningiomas are the most prevalent primary intracranial tumours, yet predicting their growth trajectories from longitudinal MRI remains an open challenge due to heterogeneous growth dynamics and limited longitudinal data. We address the question: at what level of model complexity does additional sophistication stop improving out-of-sample prediction when the longitudinal cohort contains only $N = 31$–58 patients? We organise the investigation as a three-stage complexity ladder. Stage 1 (volumetric baseline) segments tumours, extracts whole-tumour volume, and models scalar volume trajectories with GP and LME models. Stage 2 (latent severity) introduces a single patient-specific severity parameter $s_i \in [0, 1]$ via a nonlinear mixed-effects model connected to Item Response Theory and reduced Gompertz dynamics. Stage 3 (representation learning) applies BrainSegFounder—a SwinUNETR foundation model pretrained on 41,400+ subjects—adapted via Low-Rank Adaptation (LoRA), followed by a Supervised Disentangled Projection (SDP) and PCA-compressed GP prediction with ARD kernels. A cross-cutting variance decomposition quantifies the marginal ΔR^2 of each stage via Leave-One-Patient-Out Cross-Validation, with paired permutation tests and BCa bootstrap confidence intervals. Each stage must earn its place: the complexity ladder advances only when the next stage demonstrates statistically significant improvement. We report the first out-of-sample prediction benchmarks for meningioma growth, domain gap analysis for the glioma-to-meningioma transfer, LoRA ablation results, and discuss

methodology revisions motivated by empirical findings including the failure of shape disentanglement and the marginal benefit of LoRA adaptation ($\Delta R^2 = +0.016$) for downstream feature quality.

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1 Introduction

1.1 Clinical Motivation

Meningiomas are the most common primary intracranial tumours, accounting for approximately 39% of all central nervous system neoplasms, with an age-adjusted incidence rate of 9.51 per 100,000 population [1]. The majority are classified as WHO Grade I (benign), and a substantial fraction are discovered incidentally on neuroimaging performed for unrelated indications [2]. For these incidentally discovered meningiomas, clinical management hinges on a deceptively simple question: *will this tumour grow?*

Current clinical practice relies on serial MRI surveillance with manual volumetric assessment at intervals of 3–12 months [3, 4]. This approach has three fundamental limitations. First, manual volumetry is labour-intensive and subject to considerable inter-observer variability, with reported coefficient of variation exceeding 15% for small tumours [5]. Second, the decision to intervene (surgery or radiotherapy) is typically made reactively—after significant growth has already occurred—rather than proactively based on predicted trajectories. Third, meningioma growth dynamics are heterogeneous: approximately 67% show measurable growth at 4–5 year follow-up [3], but growth patterns range from stable quiescence through steady linear expansion to Gompertz-type saturation [6, 7], and no established imaging biomarker reliably predicts which pattern a given tumour will follow. Volumetric assessment substantially outperforms diameter-based measurements for detecting growth, with 15.2% sensitivity versus 5.6% using linear diameter [8].

Quantitative growth prediction from longitudinal MRI could transform meningioma management by enabling personalised surveillance schedules, early identification of aggressive growth patterns, and evidence-based timing of intervention. However, building such predictive models faces a data bottleneck: longitudinal meningioma MRI datasets are rare, typically small (tens of patients), and collected retrospectively across heterogeneous scanners and protocols.

1.2 The Core Question: Complexity versus Parsimony at Small N

The central question of this thesis is not merely "can we predict meningioma growth?" but rather: *at what level of model complexity does additional sophistication stop improving out-of-sample prediction when the longitudinal cohort contains only $N = 31$–58 patients?*

This question is both practical and methodological. On the practical side, the answer determines whether clinical growth prediction requires expensive deep learning pipelines or whether simpler segmentation-based volumetric models suffice. On the methodological side, it addresses a widespread gap in the medical imaging literature, where increasingly complex architectures are proposed without systematic comparison against simpler base-

lines under the same evaluation protocol.

A key constraint is the effective sample size. With $N = 31$–58 patients averaging approximately 3.6 observations each ($\sim$112 total observations), the maximum number of predictor parameters that can be reliably estimated is 2–3 at the most conservative rule of thumb of 15–20 subjects per predictor [9, 10]. This places a hard ceiling on model complexity: any model that exceeds this budget must earn its place through demonstrably superior out-of-sample performance.

To our knowledge, no published work provides out-of-sample prediction benchmarks for meningioma growth. Existing studies are limited to retrospective volumetric characterisation and within-sample growth curve fitting [3, 11, 6]. This thesis addresses that gap directly.

1.3 Experimental Design: A Three-Stage Complexity Ladder

Rather than proposing a single pipeline and reporting its metrics, we organise the experimental programme as a *three-stage complexity ladder* in which each stage must justify its existence through measurable improvement over the preceding one. A cross-cutting variance decomposition quantifies the marginal ΔR^2 contributed by each stage.

Stage 1 — Segmentation-Based Volumetric Baseline (Section 8.2). The simplest credible approach: segment the tumour from each MRI (using a frozen or adapted encoder), extract whole-tumour volume in mm^3, and model the resulting scalar volume trajectories with GP and LME models. This stage requires only a reliable segmentation model and a single scalar per observation. Three models of increasing flexibility are evaluated: a pooled scalar GP (ScalarGP), a linear mixed-effects model (LME), and a hierarchical GP with population mean from the LME (HGP). An optional Gompertz mean function is assessed as an alternative to the linear population trend.

Stage 2 — Latent Severity Model (Section 8.3). A nonlinear mixed-effects (NLME) model posits a single latent severity variable $s_i \in [0, 1]$ per patient, estimated via quantile transformation of growth patterns. This is formally connected to Item Response Theory (IRT): patients are "examinees" with latent ability s_i, timepoints are "items" with difficulty determined by t_{ij}, and observed growth is the "response" governed by a monotonic link function [12, 13]. Stage 2 adds at most 1–2 parameters per patient beyond Stage 1 while providing a richer individual-level characterisation.

Stage 3 — Representation Learning (Section 8.4). The full deep learning pipeline: BrainSegFounder [14] encoder adapted via LoRA [15] (Section 5), followed by a Supervised Disentangled Projection [16] (Section 6) mapping $\mathbf{h} \in \mathbb{R}^{768}$ to $\mathbf{z} \in \mathbb{R}^{128}$, with PCA-compressed latent features fed to GP models with Automatic Relevance

Determination (ARD) kernels. This is the most complex stage and the last to be attempted: it must demonstrate that high-dimensional deep features capture growth-relevant information beyond what volume and severity already explain.

The critical design principle is that **each stage must earn its place**. Stage 2 is only warranted if it improves upon Stage 1. Stage 3 is only warranted if it improves upon Stage 2. If volume alone captures the dominant growth signal—as parsimony would suggest at $N = 31$–58—then the primary scientific contribution is establishing that fact rigorously, not building a more complex model.

1.4 Contributions

This thesis makes the following contributions:

1. **The first out-of-sample meningioma growth prediction benchmark.** Using Leave-One-Patient-Out Cross-Validation (LOPO-CV) on a longitudinal cohort of 31–58 patients, we provide the first reported R^2, MAE, and calibration metrics for meningioma volume prediction. The volumetric baseline (Stage 1) establishes a reference against which all subsequent complexity can be measured.

2. **A latent severity framework for growth characterisation.** We formulate meningioma growth heterogeneity as a latent variable problem connected to IRT and reduced Gompertz dynamics [12], where a single patient-specific severity parameter s_i explains inter-patient variability. This provides interpretable patient stratification while adding minimal parameters.

3. **A systematic assessment of representation learning value.** By placing the full BrainSegFounder + LoRA + SDP pipeline as Stage 3 of the complexity ladder, we directly quantify the marginal ΔR^2 of deep features over volumetric and severity baselines. This addresses the recurring question in medical imaging of whether foundation model features improve upon simpler alternatives for specific downstream tasks.

4. **Variance decomposition as an analytical framework.** The cross-cutting analysis decomposes prediction variance across the three stages, providing not only a model selection criterion but also a methodological template for evaluating complexity–benefit tradeoffs in small-sample longitudinal prediction.

1.5 Document Structure

The remainder of this report is organised as follows. Section 2 reviews the theoretical foundations: meningioma growth biology, GP regression for longitudinal data, nonlinear

mixed-effects models, sample size constraints, and representation learning. Section 3 describes the datasets, preprocessing, and data splits. Section 4 presents the domain gap analysis motivating LoRA adaptation. Section 5 details the LoRA encoder adaptation (Phase 1, relevant to Stage 3). Section 6 describes the Supervised Disentangled Projection (Phase 2, relevant to Stage 3). Section 7 covers cohort encoding and ComBat harmonisation (Phase 3). Section 8 presents the three-stage complexity ladder for growth prediction, which is the core methodological contribution. Section 9 consolidates the variance decomposition, quality targets, and ablation results. Section 10 discusses limitations and future directions.

2 Background and Related Work

2.1 Meningioma Growth Biology

Understanding the biological dynamics of meningioma growth is essential for selecting appropriate mathematical models. Unlike gliomas, which are infiltrative parenchymal tumours, meningiomas are extra-axial masses arising from the arachnoid cap cells of the meninges. They grow by displacing rather than infiltrating brain tissue, and their dural attachment constrains their spatial dynamics.

2.1.1 Growth Patterns and Subgroups

Engelhardt et al. [6] conducted a comprehensive analysis of meningioma growth trajectories and found that the Gompertz model provides the best fit among standard parametric growth curves. They identified three growth subgroups: (i) non-growing tumours that remain volumetrically stable; (ii) slowly growing tumours with near-linear dynamics over clinical timescales; and (iii) rapidly growing tumours displaying clear deceleration consistent with Gompertz saturation. Approximately 67% of incidental meningiomas show measurable growth at 4–5 year follow-up [3].

Fountain et al. [7] compared logistic and Gompertz models for benign versus atypical meningiomas, confirming that saturating growth models outperform exponential fits for most patients. The critical clinical implication is that short-term growth rates do not reliably predict long-term behaviour: a tumour on the decelerating phase of a Gompertz curve may appear aggressive early but stabilise later.

2.1.2 Volumetric versus Diameter Measurements

Benjamin et al. [8] demonstrated that volumetric assessment is substantially more sensitive than diameter-based measurements for detecting meningioma growth, with detection

rates of 15.2% versus 5.6%. This motivates our use of whole-tumour volume as the primary growth endpoint and underscores the importance of accurate segmentation.

2.1.3 Clinical Implications for Modelling

Three biological facts constrain the modelling framework:

1. **Meningiomas do not migrate.** As extra-axial, dural-attached tumours, the spatial centroid is essentially static over time (modulo asymmetric growth). Location is therefore a patient-level covariate, not a temporal variable.

2. **Growth heterogeneity is high.** The three-subgroup structure [6] implies that population-level models must accommodate qualitatively different growth regimes, motivating mixed-effects or latent variable approaches.

3. **Observation density is low.** Typical clinical follow-up yields 2–6 scans per patient over months to years, with irregular spacing. Models must handle sparse, irregular time series gracefully.

2.2 Gaussian Process Regression for Longitudinal Data

A Gaussian process (GP) is a distribution over functions such that any finite collection of function values is jointly Gaussian [17]:

$$f(\mathbf{t}) \sim \mathcal{GP}\Big(m(\mathbf{t}),\ k(\mathbf{t}, \mathbf{t}')\Big), \tag{1}$$

where $m(\mathbf{t})$ is the mean function and $k(\mathbf{t}, \mathbf{t}')$ is the covariance (kernel) function. Given n observations $\mathbf{y} = f(\mathbf{t}) + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, the predictive distribution at test inputs $\mathbf{t}_*$ is:

$$\mathbf{f}_* | \mathbf{t}_*, \mathbf{t}, \mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*), \tag{2}$$

$$\boldsymbol{\mu}_* = m(\mathbf{t}_*) + \mathbf{K}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} (\mathbf{y} - m(\mathbf{t})), \tag{3}$$

$$\boldsymbol{\Sigma}_* = \mathbf{K}_{**} - \mathbf{K}_*^\top (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{K}_*, \tag{4}$$

where $\mathbf{K} = k(\mathbf{t}, \mathbf{t})$, $\mathbf{K}_* = k(\mathbf{t}, \mathbf{t}_*)$, and $\mathbf{K}_{**} = k(\mathbf{t}_*, \mathbf{t}_*)$.

2.2.1 Matérn Kernels

The Matérn family of kernels provides control over the smoothness of the GP sample paths via the parameter ν:

$$k_\nu(\Delta t) = \sigma_f^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{\sqrt{2\nu}\, |\Delta t|}{\ell} \right)^\nu K_\nu \left(\frac{\sqrt{2\nu}\, |\Delta t|}{\ell} \right), \tag{5}$$

where ℓ is the length-scale, σ_f^2 the signal variance, and K_ν the modified Bessel function. For $\nu = 5/2$, the kernel yields twice-differentiable sample paths:

$$k_{5/2}(\Delta t) = \sigma_f^2 \left(1 + \frac{\sqrt{5}\,|\Delta t|}{\ell} + \frac{5\,\Delta t^2}{3\ell^2}\right) \exp\left(-\frac{\sqrt{5}\,|\Delta t|}{\ell}\right). \tag{6}$$

This is our default temporal kernel for tumour volume dynamics, encoding the prior belief that volume evolves smoothly over time.

2.2.2 Connection to Linear Mixed-Effects Models

A GP with a linear kernel $k(t, t') = \sigma_0^2 + \sigma_1^2 t t'$ and zero mean is mathematically equivalent to a random-intercept, random-slope LME [17]. This connection is not merely formal: it means that the LME baseline (Stage 1) is a special case of the GP framework, and improvements from the GP must come from the nonlinear component captured by the Matérn kernel. This nesting property ensures that the GP can never perform worse than the LME on training data and provides a principled model comparison via log-marginal likelihood.

2.2.3 Hierarchical GP for Longitudinal Cohorts

When multiple patients share similar growth dynamics, a hierarchical (empirical Bayes) GP structure pools information across patients. The Gaussian Process Panel Model (GPPM) framework of Karch et al. [18] formalises this: shared kernel hyperparameters $\boldsymbol{\theta} = (\sigma_f, \ell, \sigma_n)$ are estimated from the pooled marginal likelihood across all patients, while patient-specific posteriors arise from conditioning on individual observations. This is particularly important at small N: per-patient kernel fitting with $n_i \in [2, 6]$ observations is ill-conditioned, but pooling across $N = 33$ patients with $\sim$100 total observations provides stable hyperparameter estimates.

2.2.4 Multi-Output GPs and the Intrinsic Coregionalisation Model

Multi-Output Gaussian Processes (MOGPs) extend scalar GPs to vector-valued functions $\mathbf{f} : \mathbb{R} \to \mathbb{R}^D$, capturing correlations across output dimensions. The Intrinsic Coregionalisation Model (ICM) [19] parameterises the cross-output covariance as a Kronecker product:

$$\mathbf{K}(\mathbf{t}, \mathbf{t}') = \mathbf{B} \otimes k_t(\mathbf{t}, \mathbf{t}'), \tag{7}$$

where $\mathbf{B} \in \mathbb{R}^{D \times D}$ is a positive semi-definite coregionalisation matrix encoding inter-output correlations, and k_t is a scalar temporal kernel shared across outputs. The Linear Model

of Coregionalisation (LMC) generalises this to a sum over Q latent functions:

$$\mathbf{K}(\mathbf{t}, \mathbf{t}') = \sum_{q=1}^{Q} \mathbf{B}_q \otimes k_q(\mathbf{t}, \mathbf{t}'). \tag{8}$$

2.3 Nonlinear Mixed-Effects Models and Latent Severity

2.3.1 NLME Formulation

Nonlinear mixed-effects (NLME) models extend the LME framework to nonlinear growth functions while preserving the population–individual decomposition [20]. For patient i at timepoint j:

$$y_{ij} = g(s_i, t_{ij}; \boldsymbol{\theta}) + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \tag{9}$$

where g is a nonlinear growth function, s_i is a patient-specific latent variable (the "severity" or growth propensity), and $\boldsymbol{\theta}$ are population-level parameters.

2.3.2 Reduced Gompertz Model

Vaghi et al. [12] demonstrated that for tumour growth modelling, a *reduced Gompertz* model with a single patient-specific parameter explains the majority of inter-patient growth heterogeneity:

$$V_i(t) = V_{i,0} \exp\left(\frac{\alpha_i}{\beta}\left(1 - e^{-\beta t}\right)\right), \tag{10}$$

where β is a shared deceleration parameter and α_i is the patient-specific initial growth rate. This aligns with Engelhardt et al.'s finding [6] that Gompertz dynamics best describe meningioma growth. The single-parameter reduction is essential at small N: with only $\sim$3.6 observations per patient on average, estimating more than 1–2 individual-level parameters per patient is infeasible.

2.3.3 Connection to Item Response Theory

The severity model has a formal connection to Item Response Theory (IRT) [21]. In the IRT analogy:

- Patients are "examinees" with latent ability $s_i \in [0, 1]$.
- Timepoints are "items" with difficulty determined by the elapsed time t_{ij}.
- Observed growth is the "response" governed by a monotonic link function.

Proust-Lima et al. [13, 22] developed joint latent class mixed models (JLCM) that formalise this connection for longitudinal biomarker trajectories. Our Stage 2 severity model can be viewed as a single-class JLCM with a continuous (rather than discrete) latent variable.

2.3.4 Monotonic Constraints

The severity variable s_i must induce a monotonic relationship with observed growth (higher severity implies faster growth at all times). This can be enforced through:

- **Parametric constraints:** Using growth functions that are monotonically increasing in s_i by construction (e.g., reduced Gompertz with $\alpha_i = f(s_i)$ where f is monotone).
- **Constrained monotonic neural networks (CMNN):** Runje & Shankaranarayana [23] provide architectures with provable monotonicity via non-negative weight constraints.
- **Monotonic GPs:** Riihimäki & Vehtari [24] impose derivative constraints on the GP posterior to enforce monotonicity.

2.4 Sample Size Constraints for Predictive Modelling

The longitudinal cohort in this work contains $N = 31$–58 patients with approximately 112 total observations. This places severe constraints on model complexity.

2.4.1 Subjects-per-Variable Rules

Riley et al. [9] provide sample size criteria for prediction models, recommending a minimum of 15–20 subjects per predictor parameter (events per variable, EPV) for stable coefficient estimation. Harrell [10] similarly advocates a maximum of $N/15 \approx 2$–3 predictor parameters for $N = 31$–58. These rules imply that the growth prediction model should have at most 2–3 free parameters per patient, ruling out high-dimensional neural approaches.

2.4.2 Effective Sample Size in Multilevel Models

For hierarchical/mixed-effects models, the effective sample size depends on both the number of patients N and the number of observations per patient n_i. Snijders & Bosker [25] note that level-2 parameters (between-patient effects) depend on N, not on $\sum n_i$. This is critical: even though we have $\sim$112 total observations, the effective sample size for patient-level parameters remains $N = 31$–58.

2.4.3 Cross-Validation Error at Small N

Varoquaux [26] demonstrated that cross-validation error estimates have standard deviations of approximately $\pm 10\%$ at $N = 100$. At $N = 33$, the uncertainty is larger still. This motivates our use of LOPO-CV (maximising training set size) and bootstrap confidence intervals, while cautioning against over-interpreting small differences between models.

2.5 Foundation Models for Brain MRI

The paradigm of pretraining large encoder–decoder architectures on massive, diverse datasets and subsequently fine-tuning on downstream tasks has transformed medical image analysis [27, 28]. BrainSegFounder [14] instantiates this paradigm for 3D brain MRI segmentation through a three-stage training pipeline:

1. **Self-supervised pretraining.** A SwinUNETR encoder [29, 30] is pretrained on 41,400+ brain MRI subjects from the UK Biobank using a combination of masked image modelling, contrastive learning, and rotation prediction.

2. **Supervised pretraining.** The encoder is paired with the SwinUNETR decoder and trained on the BraTS 2021 glioma segmentation benchmark [31] at 128^3 ROI.

3. **Per-fold fine-tuning.** Final fine-tuning on BraTS 2021 with cross-validation yields the released checkpoints.

2.5.1 SwinUNETR Architecture

SwinUNETR combines the Swin Transformer [32] with a UNet-style encoder–decoder structure. The encoder consists of a patch embedding layer followed by four transformer stages with shifted window self-attention. For BSF-Tiny (the variant used in this work):

Table 1: SwinUNETR encoder (BSF-Tiny) architecture. Input: $[B, 4, 128^3]$.

Stage	Output shape	Channels	Heads	Blocks
Patch embed	$[B, 48, 64^3]$	48	—	—
Stage 1	$[B, 96, 32^3]$	96	3	2
Stage 2	$[B, 192, 16^3]$	192	6	2
Stage 3	$[B, 384, 8^3]$	384	12	2
Stage 4	$[B, 768, 4^3]$	768	24	2
encoder10	$[B, 768, 4^3]$	768	—	—

Global Average Pooling (GAP) over the spatial dimensions of `encoder10` produces the feature vector $\mathbf{h} \in \mathbb{R}^{768}$ used in Stage 3 of our pipeline.

2.6 Low-Rank Adaptation

Low-Rank Adaptation (LoRA) [15] provides parameter-efficient fine-tuning by injecting trainable rank-r decompositions into frozen pretrained weight matrices. For a pretrained weight matrix $\mathbf{W}_0 \in \mathbb{R}^{d\times k}$, LoRA parameterises the adapted weight as:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \quad \mathbf{B} \in \mathbb{R}^{d\times r},\ \mathbf{A} \in \mathbb{R}^{r\times k}, \tag{11}$$

where $r \ll \min(d, k)$ is the rank, α is a scaling hyperparameter, $\mathbf{A}$ is initialised from $\mathcal{N}(0, \sigma^2)$, and $\mathbf{B} = \mathbf{0}$ so that the adapted model initially reproduces the pretrained model exactly.

DoRA [33] extends LoRA by decomposing the weight update into magnitude and direction components:

$$\mathbf{W} = m \cdot \frac{\mathbf{W}_0 + \mathbf{BA}}{\|\mathbf{W}_0 + \mathbf{BA}\|_c}, \tag{12}$$

where m is a learnable magnitude vector and $\| \cdot \|_c$ denotes column-wise normalisation. We evaluate DoRA as an ablation condition (A5, Section 9.3).

In the context of our three-stage ladder, LoRA adaptation is relevant to Stage 3 (representation learning). It is also relevant to Stage 1 insofar as an adapted encoder may produce better segmentations and hence more accurate volume estimates. We evaluate this explicitly in ablation A1 (Section 9.3).

2.7 Dimensionality Reduction for Deep Features

The BrainSegFounder encoder produces 768-dimensional feature vectors, which must be compressed for use in growth prediction models at small N.

2.7.1 PCA with Supervised Selection

Principal Component Analysis provides an unsupervised linear projection to a lower-dimensional subspace. For Stage 3, we apply PCA within each LOPO-CV fold (to prevent information leakage) and select 3–6 components, guided by: (i) the elbow of the variance-explained curve; (ii) correlation with the volume growth target; and (iii) the sample size constraint of 2–3 parameters per predictor.

2.7.2 ARD Kernels

Automatic Relevance Determination (ARD) kernels assign a separate length-scale ℓ_d per input dimension [17]. During marginal likelihood optimisation, irrelevant dimensions acquire large length-scales (effectively deactivating them). ARD provides an alternative to explicit dimensionality reduction, performing implicit feature selection within the GP. However, at $N = 33$ the number of ARD hyperparameters must be carefully controlled to avoid overfitting.

2.7.3 Active Subspaces

Tripathy et al. [34] formalise the notion of active subspaces for dimensionality reduction in GP regression, identifying the low-dimensional linear subspace of the input that captures most of the variation in the function output. This provides a principled alternative to PCA

that accounts for the target variable, and is conceptually related to the SDP's supervised projection in Stage 3.

2.8 Disentangled Representation Learning

The impossibility theorem of Locatello et al. [16] established that unsupervised disentanglement is theoretically impossible without inductive biases. Their follow-up work [35] demonstrated that as few as 100 labelled examples suffice to resolve this impossibility through supervision.

These results motivate our Supervised Disentangled Projection (SDP) approach in Stage 3: rather than relying on unsupervised disentanglement, we use explicit semantic supervision from segmentation-derived features to anchor each latent partition to a known generative factor. The SDP combines semantic regression, VICReg-style [36] variance–covariance regularisation, and distance correlation [37] penalties. Spectral normalisation [38] on all linear layers bounds the global Lipschitz constant, ensuring that encoder feature noise is not amplified through the projection [39]. Full details are in Section 6.

2.9 Related Work

2.9.1 Brain Tumour Growth Prediction

Tumour growth prediction from longitudinal MRI has been addressed through several paradigms. Classical approaches fit parametric growth models (exponential, Gompertz, logistic) to volume trajectories [40, 3]. Reaction–diffusion models [41, 42] incorporate spatial diffusion but require tissue-specific parameterisation and are computationally expensive.

Recent deep learning approaches include spatio-temporal convolutional LSTMs [43] achieving 83.2% Dice on glioma growth prediction, GP-GAN [44] combining Gaussian processes with GANs, and neural fields conditioned on time [45] for vestibular schwannoma growth. Puglisi et al. [46] demonstrated that a linear mixed-effects model in a learned latent space produces clinically meaningful progression predictions for brain atrophy, trained on 11,730 MRIs.

For meningiomas specifically, most existing work is limited to retrospective volumetric analysis [3, 4, 11]. To our knowledge, no prior work has provided out-of-sample prediction benchmarks for meningioma growth.

2.9.2 Latent Space Temporal Prediction

The Joint Embedding Predictive Architecture (JEPA) framework [47] advocates predicting in representation space rather than pixel space. Puglisi et al. [46] validated this principle for brain atrophy using LME in a learned latent space. Our Stage 3 aligns

with this "predict in latent space" principle while our Stages 1 and 2 serve as necessary baselines to determine whether the latent space provides value beyond simpler volumetric features.

2.9.3 Domain Adaptation in Medical Imaging

Domain adaptation for medical image analysis has been addressed through adversarial training [48], style transfer [49], and parameter-efficient fine-tuning of foundation models [50]. Ben-David et al. [51] established the theoretical framework relating domain divergence to target risk. Our LoRA adaptation (Stage 3) contributes empirical evidence on the marginal value of domain adaptation when a strong SSL-pretrained encoder is available.

3 Data

This section describes the three datasets, the unified HDF5 storage schema, data splits, and the preprocessing pipeline.

3.1 Datasets

3.1.1 BraTS-MEN 2024 (Cross-Sectional Meningioma)

The BraTS Meningioma 2024 challenge dataset [52] comprises 1,000 subjects with pre-operative multi-parametric MRI (T1, T1ce, T2, FLAIR) and expert manual segmentations. Each subject has a single study (timepoint). Segmentation labels follow the BraTS convention: 0 (background), 1 (enhancing tumour, ET), 2 (non-enhancing tumour, NET), and 3 (cystic component). This dataset serves as the primary training source for Phases 1 and 2.

3.1.2 BraTS-GLI 2021 (Longitudinal Glioma)

The BraTS Glioma 2021 challenge dataset [31] contains 1,350 scans from 613 patients (1–10 timepoints per patient) with the same four MRI modalities. Glioma labels use: 0 (background), 1 (necrotic tumour core, NETC), 2 (peritumoral oedema, SNFH), 3 (enhancing tumour, ET), and 4 (resection cavity, RC). This dataset is used for domain gap analysis (Section 4) and dual-domain LoRA training (Section 5.6).

3.1.3 Andalusian Longitudinal Cohort (MenGrowth)

A retrospective longitudinal cohort of 42 meningioma patients (137 total studies) collected across hospitals in Andalusia, Spain. After quality control, 33 patients (100 studies) with 2–6 timepoints each are retained for growth prediction. Observation statistics are summarised in Table 2.

Table 2: Andalusian longitudinal cohort statistics (after QC).

Statistic	**Value**
Total patients (N)	33
Total studies (S)	100
Mean studies/patient ($\bar{n}$)	3.03
Std studies/patient (σ_n)	1.17
Range	[2, 6]
Patients with exactly 2 studies	19 (57.6%)

3.2 Unified HDF5 Schema

All three datasets are stored in a unified HDF5 v2.0 schema, enabling a single reader code path (`BraTSDatasetH5`) regardless of dataset type. The schema structure is:

Table 3: Unified HDF5 v2.0 schema. N_s: number of scans, N_p: number of patients.

Key	**Shape/Type**	**Description**
`attrs`	dict	{`n_scans`, `n_patients`, `roi_size`, `spacing`, `channel_order`, `version`="2.0", `dataset_type`, `domain`}
`images`	$[N_s, 4, 192^3]$ float32	Multi-modal MRI volumes
`segs`	$[N_s, 1, 192^3]$ int8	Segmentation masks
`scan_ids`	$[N_s]$ str	Unique scan identifiers
`patient_ids`	$[N_s]$ str	Patient identifiers
`timepoint_idx`	$[N_s]$ int32	Per-patient timepoint index
`semantic/`	group	Precomputed features: `volume` $[N_s, 4]$, `location` $[N_s, 3]$, `shape` $[N_s, 3]$
`longitudinal/`	group	CSR structure: `patient_offsets` $[N_p + 1]$, `patient_list` $[N_p]$
`splits/`	group	Patient-level index arrays: `lora_train`, `lora_val`, `test`
`metadata/`	group	`grade`, `age`, `sex` (when available)

For cross-sectional data (BraTS-MEN), the longitudinal structure is trivial: each patient has exactly one scan at timepoint 0, so $N_s = N_p$ and `patient_offsets` $= [0, 1, 2, \ldots, N]$. This design ensures that all timepoints of a patient are assigned to the same split, preventing patient-level data leakage.

3.3 Data Splits

Data splitting is performed at the patient level with seed 42 for reproducibility. Table 4 summarises the splits for each dataset.

Table 4: Data splits. BraTS-MEN splits are canonical; server configurations may differ slightly (see Section 5.7).

Split	BraTS-MEN	BraTS-GLI (patients)	MenGrowth
`lora_train`	750	430 (∼950 scans)	23
`lora_val`	100	60 (∼130 scans)	3
`test`	150	123 (∼270 scans)	7
Total	1,000	613 (1,350 scans)	33 (100 scans)

For Phase 2 (SDP), the canonical specification defines an additional `sdp_train` split of 225 subjects; however, the actual implementation uses the `lora_train` pool since the SDP operates on frozen encoder features and there is no gradient contamination concern.

3.4 Preprocessing Pipeline

All volumes are spatially preprocessed to 192^3 voxels at $1\,\text{mm}^3$ isotropic spacing during HDF5 conversion. Intensity normalisation is performed at runtime. The critical channel ordering convention is $[\text{FLAIR}, \text{T1ce}, \text{T1}, \text{T2}]$, matching BrainSegFounder's training protocol—incorrect ordering causes near-zero Dice (~ 0.00) even on the glioma training domain.

3.4.1 Training Transforms

The training pipeline follows the BrainSegFounder protocol with minor adaptations:

1. `CropForegroundd`: Crop to the non-zero brain region with `k_divisible`$= [128, 128, 128]$.
2. `SpatialPadd`: Pad to at least 128^3.
3. `RandSpatialCropd`: Random crop to 128^3.
4. `NormalizeIntensityd`: Per-channel, nonzero-voxel normalisation.
5. Data augmentation: random flips (3 axes, $p = 0.5$ each), intensity scaling (factor 0.1, $p = 1.0$), intensity shift (offset 0.1, $p = 1.0$).

3.4.2 Feature Extraction Transforms

For feature extraction and evaluation, a larger 192^3 ROI is used (via `ResizeWithPadOrCropd`) to ensure 100% tumour containment—the 128^3 centre crop captures only 38.8% of meningioma tumours fully due to their dural-based, peripheral location.

3.4.3 Inference Transforms

Full-resolution inference uses MONAI's sliding window inference with 128^3 patches, 50% overlap, and Gaussian blending.

3.5 Semantic Features

Semantic features are precomputed from segmentation masks and stored in the HDF5 files. Three categories of features are extracted:

Volume (4 dims): $[\log(V_{\text{total}} + 1), \log(V_{\text{NCR}} + 1), \log(V_{\text{ED}} + 1), \log(V_{\text{ET}} + 1)]$, where volumes are in mm^3. The log-transform stabilises the heavy-tailed distribution.

Location (3 dims): Centroid coordinates (c_z, c_y, c_x) in mm, computed as the centre of mass of the tumour segmentation mask.

Shape (3 dims): Sphericity $\in (0, 1]$, log(surface area), and solidity (ratio of tumour volume to convex hull volume), computed via `compute_shape_array()`.

Remark 1. *Following the R1 methodology revision (Section 9.6), the volume target is reduced to a single whole-tumour log-volume* $v^* = \log(V_{WT} + 1) \in \mathbb{R}^1$*, and shape features are dropped from the SDP due to persistent low* R^2 *(*≤ 0.11 *across all conditions).*

3.6 Segmentation Label Conversion

Both datasets are converted to a unified 3-channel hierarchical overlapping representation for the segmentation loss:

$$\text{TC} = \{x : \text{label}(x) \in S_{\text{TC}}\}, \tag{13}$$

$$\text{WT} = \{x : \text{label}(x) > 0\}, \tag{14}$$

$$\text{ET} = \{x : \text{label}(x) \in S_{\text{ET}}\}, \tag{15}$$

where S_{TC} and S_{ET} are domain-dependent. For MEN: $S_{\text{TC}} = \{1, 2\}$, $S_{\text{ET}} = \{1\}$. For GLI: $S_{\text{TC}} = \{1, 3, 4\}$, $S_{\text{ET}} = \{3\}$. This conversion is implemented in `segmentation.py._convert_target()` and ensures that the segmentation decoder output semantics are consistent across domains.

4 Domain Gap Analysis

BrainSegFounder was pretrained and fine-tuned on glioma data. Before adapting it to the meningioma domain, we quantify the distributional shift in the frozen encoder's feature space to establish the empirical necessity of domain adaptation.

4.1 Feature Extraction Protocol

Features are extracted from the frozen BrainSegFounder encoder using the `FeatureExtractor` pipeline:

$$\mathbf{h} = \mathrm{GAP}\Big(\texttt{encoder10}\Big(\mathrm{SwinViT}(\mathbf{x})\Big)\Big) \in \mathbb{R}^{768}, \tag{16}$$

where $\mathbf{x} \in \mathbb{R}^{4\times 192^3}$ is the input volume and GAP denotes Global Average Pooling over the 4^3 (or 6^3 for 192^3 input) spatial dimensions. We extract features from 200 random BraTS-GLI subjects and 200 BraTS-MEN subjects from the test and SDP-train splits.

4.2 Domain Gap Metrics

4.2.1 Maximum Mean Discrepancy

The Maximum Mean Discrepancy (MMD) [53] measures the distance between two distributions in a reproducing kernel Hilbert space. Given samples $\{\mathbf{h}_i^{\mathrm{GLI}}\}_{i=1}^n$ and $\{\mathbf{h}_j^{\mathrm{MEN}}\}_{j=1}^m$, the unbiased estimator is:

$$\widehat{\mathrm{MMD}}^2 = \frac{1}{n(n-1)} \sum_{i\neq i'} k(\mathbf{h}_i, \mathbf{h}_{i'}) + \frac{1}{m(m-1)} \sum_{j\neq j'} k(\mathbf{h}_j, \mathbf{h}_{j'}) - \frac{2}{nm} \sum_{i,j} k(\mathbf{h}_i, \mathbf{h}_j), \tag{17}$$

where k is a Gaussian kernel with bandwidth σ set to the median pairwise distance (the median heuristic). Statistical significance is assessed via a permutation test with $n_{\mathrm{perm}} = 1{,}000$ permutations.

4.2.2 Centred Kernel Alignment

CKA [54] measures representational similarity between two feature matrices $\mathbf{H}_1 \in \mathbb{R}^{n\times d_1}$ and $\mathbf{H}_2 \in \mathbb{R}^{n\times d_2}$ via:

$$\mathrm{CKA}(\mathbf{H}_1, \mathbf{H}_2) = \frac{\|\bar{\mathbf{H}}_1^\top \bar{\mathbf{H}}_2\|_F^2}{\|\bar{\mathbf{H}}_1^\top \bar{\mathbf{H}}_1\|_F \cdot \|\bar{\mathbf{H}}_2^\top \bar{\mathbf{H}}_2\|_F}, \tag{18}$$

where $\bar{\mathbf{H}}$ denotes column-centred matrices. CKA $\in [0, 1]$, with 1 indicating identical representations (up to linear transformation).

4.2.3 Linear Probes

Following Alain and Bengio [55], we train Ridge regression probes from frozen features $\mathbf{h}$ to semantic targets (volume, location, shape). Cross-validated R^2 quantifies the linear decodability of each semantic factor. Low R^2 on the target domain indicates that the frozen encoder has not learned meningioma-relevant features.

4.3 Results

Table 5: Domain gap metrics between frozen BrainSegFounder features on BraTS-GLI vs. BraTS-MEN. Results confirm significant distributional shift motivating domain adaptation.

Metric	**Value**	**Interpretation**
MMD^2 (GLI vs. MEN)	*[to be filled]*	> 0 confirms shift
MMD p-value (permutation test)	*[to be filled]*	< 0.05 confirms significance
CKA (GLI vs. MEN)	*[to be filled]*	Low $\Rightarrow$ different structure
Linear probe R^2 (volume, MEN)	*[to be filled]*	Baseline decodability
Linear probe R^2 (location, MEN)	*[to be filled]*	Baseline decodability
Linear probe R^2 (shape, MEN)	*[to be filled]*	Baseline decodability

Figure placeholder: UMAP of frozen BrainSegFounder features. GLI (orange) vs. MEN (blue). Expected: clear separation.

Figure 1: UMAP visualisation of frozen BrainSegFounder encoder features ($\mathbf{h} \in \mathbb{R}^{768}$) for BraTS-GLI (orange, $n = 200$) and BraTS-MEN (blue, $n = 200$). The visible separation confirms the domain gap motivating LoRA adaptation.

The domain gap analysis confirms that the frozen glioma-trained encoder produces features with significant distributional shift when applied to meningioma data. The MMD permutation test rejects the null hypothesis of identical distributions, and the linear probe R^2 values on MEN data are substantially lower than within-domain GLI probes, indicating that the frozen features inadequately capture meningioma-specific semantic content. These findings motivate the LoRA adaptation described in Section 5.

4.4 Effective Rank Analysis

The effective rank of the feature matrix, defined as the exponential of the spectral entropy:

$$\mathrm{erank}(\mathbf{H}) = \exp\left(-\sum_{i=1}^{d} \bar{s}_i \log \bar{s}_i\right), \quad \bar{s}_i = \frac{s_i}{\sum_j s_j}, \tag{19}$$

where s_i are the singular values of $\mathbf{H}$, measures the distributional uniformity of the representation. The frozen BrainSegFounder encoder achieves an effective rank of approximately 47.6 on MEN data, indicating moderate utilisation of the 768-dimensional feature space.

5 Phase 1: LoRA Encoder Adaptation

5.1 Adaptation Strategy

We adapt the frozen BrainSegFounder encoder to the meningioma domain via LoRA (Section 2.6), injecting low-rank adapter matrices into the self-attention projections of Stages 3–4. Stages 0–2 remain frozen to preserve low-level anatomical features learned from the UK Biobank SSL pretraining. Segmentation on BraTS-MEN 2024 serves as the proxy task.

5.1.1 Target Modules

LoRA adapters are injected into the **Q**, **K**, **V** projection matrices within the multi-head self-attention blocks of Stages 3 and 4:

$$\texttt{swinViT.layers3.0.blocks.\{0,1\}.attn.qkv.weight}, \tag{20}$$

$$\texttt{swinViT.layers4.0.blocks.\{0,1\}.attn.qkv.weight}. \tag{21}$$

This yields 4 adapted layers, each with rank-r decompositions for Q, K, and V (combined as a single QKV projection in the Swin implementation). The total number of trainable LoRA parameters is approximately 197K for $r = 8$, compared to 62M total model parameters ($< 0.32\%$).

5.1.2 LoRA Hyperparameters

Table 6: LoRA configuration (selected via rank ablation, see Section 5.5).

Parameter	Value	Rationale
Rank (r)	8	Optimal from ablation over $r \in \{2, 4, 8, 16, 32\}$
Scale (α)	16	Effective scale $\alpha/r = 2.0$
Dropout	0.1	Regularisation
Target stages	3, 4	High-level features; stages 0–2 frozen
Target modules	QKV	All attention projections

5.2 Auxiliary Semantic Heads

To enrich the encoder's feature space for downstream SDP training, we attach auxiliary semantic regression heads that predict tumour properties directly from GAP-pooled encoder features:

$$\hat{\mathbf{y}}_p = \pi_p\Big(\mathrm{GAP}(\texttt{encoder10}(\mathbf{x}))\Big), \quad p \in \{\mathrm{vol}\}, \tag{22}$$

where $\pi_{\mathrm{vol}} : \mathbb{R}^{768} \to \mathbb{R}^1$ is a linear head predicting log whole-tumour volume. The auxiliary loss is:

$$\mathcal{L}_{\mathrm{aux}} = \lambda_{\mathrm{aux}} \cdot \|\hat{\mathbf{y}}_{\mathrm{vol}} - \mathbf{y}_{\mathrm{vol}}\|_2^2. \tag{23}$$

Following the R1 revision (Section 9.6), auxiliary heads for location and shape have been removed; only the volume head is retained. The auxiliary loss is activated after epoch 5 with a linear warmup over 10 epochs ($\lambda_{\mathrm{aux}} = 0.1$ at full strength).

Remark 2. *The auxiliary heads are discarded after Phase 1. Their sole purpose is to provide multi-task learning signal during encoder adaptation, encouraging the encoder to produce features from which tumour volume can be linearly decoded.*

5.3 Training Protocol

Table 7: Phase 1 training configuration.

Parameter	Value	Rationale
Epochs	150	With early stopping (patience 25)
LR (LoRA)	10^{-4}	Standard for LoRA adaptation
LR (decoder)	5×10^{-4}	Larger for pretrained decoder
LR (aux heads)	10^{-3}	Small heads, fast convergence
Optimizer	AdamW	Decoupled weight decay
Weight decay	10^{-5}	Light regularisation
Scheduler	Cosine decay, 10-epoch warmup	Smooth convergence
Batch size	2 (per GPU)	Memory-constrained by 128^3 input
Gradient accumulation	4 steps	Effective batch size 8
Gradient clipping	$\|\mathbf{g}\|_2 \leq 1.0$	Stability
Precision	bf16-mixed	Memory efficiency on A100

The segmentation loss combines Dice loss and cross-entropy with equal weight:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{dice}}\, \mathcal{L}_{\text{Dice}} + \lambda_{\text{CE}}\, \mathcal{L}_{\text{CE}}, \quad \lambda_{\text{dice}} = \lambda_{\text{CE}} = 1.0. \tag{24}$$

The total training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + w(e)\, \mathcal{L}_{\text{aux}}, \tag{25}$$

where $w(e) = \lambda_{\text{aux}} \cdot \min(1, \max(0, (e-5)/10))$ is the warmup schedule for the auxiliary loss starting at epoch 5.

5.4 Post-Training: LoRA Merge and Checkpoint Export

After training, LoRA weights are absorbed into the base model weights:

$$\mathbf{W}_{\text{merged}} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}. \tag{26}$$

The segmentation decoder and auxiliary heads are then discarded, yielding a merged encoder checkpoint (`phase1_encoder_merged.pt`) containing only the SwinViT and encoder1–10 weights. All encoder parameters are frozen for subsequent phases.

5.5 Ablation Results

5.5.1 Rank Sweep (A1)

We evaluate LoRA ranks $r \in \{2, 4, 8, 16, 32\}$ on segmentation Dice and linear probe R^2 for volume prediction.

Table 8: LoRA rank ablation (A1). Dice scores on `lora_val` and linear probe R^2 for volume on test features.

Rank	**LoRA params**	**Dice (WT)**	**Dice (TC)**	**Dice (ET)**	**Vol R^2**
Frozen	0	*[baseline]*	*[baseline]*	*[baseline]*	*[baseline]*
2	∼49K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
4	∼99K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
8	∼197K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
16	∼394K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
32	∼787K	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*

Figure placeholder: LoRA rank ablation curves.
Left: Dice vs. rank. Right: Probe R^2 vs. rank.

Figure 2: LoRA rank ablation (A1). Left panel shows validation Dice scores per sub-region as a function of LoRA rank. Right panel shows linear probe R^2 for volume prediction. Rank $r = 8$ achieves the best trade-off between capacity and parameter efficiency.

5.5.2 LoRA vs. DoRA (A2)

DoRA (Equation (12)) is evaluated as an alternative adapter type at $r = 8$.

Table 9: LoRA vs. DoRA comparison (A2) at rank 8.

Adapter	**Dice (WT)**	**Vol R^2 (linear)**	**Vol R^2 (RBF)**
LoRA ($r = 8$)	*[to fill]*	*[to fill]*	*[to fill]*
DoRA ($r = 8$)	*[to fill]*	*[to fill]*	*[to fill]*

5.6 Dual-Domain Adaptation

A key scientific question is whether a single LoRA-adapted encoder can produce semantically structured features for both glioma and meningioma simultaneously. Dual-domain training combines BraTS-MEN (∼750 studies) and BraTS-GLI (∼950 scans) via a `ConcatDataset` with domain-balanced `WeightedRandomSampler` (50/50 domain weight).

5.6.1 Empirical Findings

Table 10: Single-domain vs. dual-domain LoRA adaptation results.

	MEN Dice	**Vol R^2**	**CV Vol R^2**	**MMD^2**	**CKA**	**erank**
Frozen baseline	0.52	0.56	—	0.118	0.002	47.6
MEN LoRA r=8	0.48	0.57	—	0.116	0.002	45.8
Dual LoRA r=8	0.57	0.62	0.712	0.037	0.017	25.4

The dual-domain encoder achieves the best volume probe R^2 and reduces the domain gap (MMD^2: 0.118 → 0.037, a 68.6% reduction), but at the cost of a substantial drop in effective rank (47.6 → 25.4, a 46.6% reduction), indicating representation compression.

5.6.2 Critical Finding: Cross-Domain Transfer Failure

Despite the reduced domain gap metrics, cross-domain GP probes reveal a critical limitation: training probes on GLI features and evaluating on MEN features yields $R^2 \approx -0.01$ for volume, indicating that the shared feature space does not support cross-domain semantic transfer. Furthermore, MEN segmentation Dice degrades from 0.64 to 0.36 under dual training, indicating catastrophic interference rather than beneficial transfer.

Remark 3. *The dual-domain results provide important negative evidence: domain gap reduction (measured by MMD, CKA) does not guarantee cross-domain semantic compatibility. The effective rank drop from 47.6 to 25.2 suggests that the encoder resolves the domain conflict by compressing both domains into a lower-dimensional shared subspace that loses domain-specific semantic content.*

5.6.3 LoRA Marginality

A surprising finding is the marginal benefit of LoRA adaptation for semantic probes: the frozen baseline achieves volume $R^2 = 0.794$, while the best LoRA condition improves this by only $\Delta R^2 = +0.016$ (within noise). This suggests that BrainSegFounder's SSL pretraining already produces features with reasonable meningioma volume decodability, and the primary benefit of LoRA is in segmentation quality rather than downstream feature quality.

5.7 Configuration

All LoRA experiments are configured via YAML files under `experiments/lora/config/`. Server configurations use 250/50/200/500 splits (lora_train/lora_val/probe_train/test), differing slightly from the canonical specification. Four conditions are defined: LoRA/-DoRA × semantic/no-semantic heads.

6 Phase 2: Supervised Disentangled Projection

6.1 Motivation

The encoder features $\mathbf{h} \in \mathbb{R}^{768}$ are high-dimensional and unstructured: semantic factors (volume, location, shape) are entangled across all 768 dimensions. Downstream GP-based growth prediction requires features that are (i) *semantically interpretable*: tumour volume should be decodable from a known subset of latent dimensions; (ii) *disentangled*: volume, location, and shape should occupy independent subspaces; (iii) *low-dimensional*: the GP covariance matrix scales as $O(D^2)$ in the output dimensionality D.

The Supervised Disentangled Projection (SDP) addresses all three requirements through a lightweight MLP trained on frozen encoder features with a composite loss combining semantic supervision, collapse prevention, and independence regularisation.

6.1.1 Why Not a VAE?

Early experiments with a Semi-Supervised VAE (SemiVAE) were abandoned due to three failure modes: (i) posterior collapse, where latent dimensions become uninformative despite extensive hyperparameter tuning (7-phase curriculum across 600 epochs); (ii) KL distortion, where the $D_{\mathrm{KL}}(q\|p)$ penalty actively minimises the mutual information $I(x; z)$ between input and representation, destroying the rich foundation model features; and (iii) reconstruction burden, where the decoder wastes capacity on pixel-level fidelity irrelevant to downstream growth prediction.

The SDP replaces the generative objective with a discriminative one that directly optimises for the properties needed by the GP temporal model, following the theoretical mandate of Locatello et al. [16]: supervised disentanglement resolves the impossibility of unsupervised approaches.

6.2 Architecture

The SDP consists of a projection network g_θ and per-partition semantic heads $\{\pi_p\}$:

$$g_\theta : \mathbb{R}^{768} \to \mathbb{R}^{128}, \quad g_\theta(\mathbf{h}) = \mathrm{SN}(\mathbf{W}_2)\, \sigma\Big(\mathrm{SN}(\mathbf{W}_1)\, \mathrm{LN}(\mathbf{h}) + \mathbf{b}_1\Big) + \mathbf{b}_2, \tag{27}$$

where LN is LayerNorm, σ is GELU with dropout $p = 0.1$, and SN$(\cdot)$ denotes spectral normalisation applied to *both* linear layers (Section 2.8). The architecture is:

$$\underbrace{\mathbf{h} \in \mathbb{R}^{768}}_{\text{frozen encoder}} \xrightarrow{\text{LN}} \xrightarrow{\text{SN-Linear(768,512)}} \xrightarrow{\text{GELU + Drop}} \xrightarrow{\text{SN-Linear(512,128)}} \underbrace{\mathbf{z} \in \mathbb{R}^{128}}_{\text{structured latent}} \tag{28}$$

The total parameter count is approximately 459K ($768 \times 512 + 512 + 512 \times 128 + 128$).

6.3 Latent Partition Structure

The 128-dimensional latent vector is partitioned into semantically supervised and unsupervised subspaces:

Table 11: Original SDP partition structure ($3 + 1$ partitions). See Section 9.6 for the R1-revised $1 + 1$ structure.

Partition	**Dims**	**Indices**	**Target dim**	**Semantic head**
$\mathbf{z}_{\text{vol}}$ (volume)	24	$[0, 24)$	4	$\pi_{\text{vol}} : \mathbb{R}^{24} \to \mathbb{R}^4$
$\mathbf{z}_{\text{loc}}$ (location)	8	$[24, 32)$	3	$\pi_{\text{loc}} : \mathbb{R}^8 \to \mathbb{R}^3$
$\mathbf{z}_{\text{shape}}$ (shape)	12	$[32, 44)$	6	$\pi_{\text{shape}} : \mathbb{R}^{12} \to \mathbb{R}^6$
$\mathbf{z}_{\text{res}}$ (residual)	84	$[44, 128)$	—	None (unsupervised)
Total	128	$[0, 128)$	13	—

Each supervised partition p has a linear semantic head $\pi_p : \mathbb{R}^{d_p} \to \mathbb{R}^{k_p}$ that predicts ground-truth semantic features extracted from segmentation masks. The residual partition captures information not explained by the three supervised factors.

Remark 4 (R1 Revision)**.** *Following empirical results (shape $R^2 \leq 0.11$, temporally static meningioma location), the partition structure is revised to $\mathbf{z} = [\mathbf{z}_{vol} \in \mathbb{R}^{32} \mid \mathbf{z}_{res} \in \mathbb{R}^{96}]$ with a single volume head $\pi_{vol} : \mathbb{R}^{32} \to \mathbb{R}^1$ predicting $\log(V_{WT} + 1)$. Location is relocated to a static GP covariate (Section 9.6).*

6.4 Composite Loss Function

The SDP training objective combines four terms:

$$\mathcal{L}_{\text{SDP}} = \underbrace{\sum_{p \in \mathcal{P}} \lambda_p \, \|\hat{\mathbf{y}}_p - \mathbf{y}_p\|_2^2}_{\text{Semantic regression}} + \underbrace{\lambda_{\text{var}} \, \mathcal{L}_{\text{var}}(\mathbf{z})}_{\text{Collapse prevention}} + \underbrace{\lambda_{\text{cov}} \, \mathcal{L}_{\text{cov}}(\mathbf{z})}_{\text{Linear independence}} + \underbrace{\lambda_{\text{dCor}} \sum_{\substack{i,j \in \mathcal{P} \\ i<j}} \text{dCor}(\mathbf{z}_i, \mathbf{z}_j)}_{\text{Nonlinear independence}}, \tag{29}$$

where $\mathcal{P}$ is the set of supervised partitions, $\hat{\mathbf{y}}_p = \pi_p(\mathbf{z}_p)$, and each loss term serves a distinct purpose.

6.4.1 Semantic Regression

For each supervised partition, MSE loss anchors the partition's information content to a known semantic factor:

$$\mathcal{L}_p^{\text{sem}} = \frac{1}{k_p} \|\pi_p(\mathbf{z}_p) - \mathbf{y}_p\|_2^2. \tag{30}$$

Targets $\mathbf{y}_p$ are normalised to zero mean and unit variance using statistics computed on the training pool only (Section 6.7).

6.4.2 Variance Hinge Loss (Collapse Prevention)

The VICReg variance term (**??**) is applied to all 128 dimensions of $\mathbf{z}$, ensuring no dimension collapses to a constant. The threshold $\gamma = 1.0$ targets unit standard deviation per dimension.

6.4.3 Covariance Penalty (Linear Independence)

Cross-partition covariance is penalised:

$$\mathcal{L}_{\text{cov}} = \frac{1}{d^2} \sum_{\substack{i,j: \\ \text{part}(i) \neq \text{part}(j)}} C_{ij}^2, \tag{31}$$

where $\mathbf{C}$ is the batch covariance matrix and $\text{part}(i)$ denotes the partition containing dimension i. Only cross-partition off-diagonal elements are penalised; within-partition correlations are permitted.

6.4.4 Distance Correlation (Nonlinear Independence)

For each pair of supervised partitions (i, j), the distance correlation $\text{dCor}(\mathbf{z}_i, \mathbf{z}_j)$ (**??**) captures nonlinear dependencies not detected by the covariance penalty. The empirical distance correlation is computed from $O(N^2)$ pairwise distance matrices, which for $N = 800$ (full-batch training) requires approximately 30 MB of GPU memory per partition pair.

6.5 Curriculum Training Schedule

Following the capacity-annealing principle from the disentanglement literature [56], losses are activated in a staged curriculum:

Table 12: SDP curriculum training schedule (4 phases).

Phase	Epochs	Active losses
Warm-up	0–9	$\mathcal{L}_{\text{var}}$ only
Semantic	10–39	$+\mathcal{L}_{\text{vol}}, \mathcal{L}_{\text{loc}}, \mathcal{L}_{\text{shape}}$
Independence	40–59	$+\mathcal{L}_{\text{cov}}, \mathcal{L}_{\text{dCor}}$
Full	60–100	All losses at full strength

The rationale is that semantic regression must first establish the partition structure before independence regularisation is imposed. Premature independence penalties can collapse dimensions before they have acquired semantic content.

6.6 Hyperparameters

Table 13: SDP training hyperparameters.

Parameter	Value	Rationale
λ_{vol}	20.0	Highest weight; volume is primary target
λ_{loc}	12.0	Moderate weight
λ_{shape}	15.0	Higher than location (harder to learn)
λ_{cov}	5.0	Cross-partition linear independence
λ_{var}	5.0	Collapse prevention
λ_{dCor}	2.0	Nonlinear independence
Learning rate	10^{-3}	Standard for small MLPs
Optimizer	AdamW	Decoupled weight decay
Weight decay	0.01	Regularisation
Epochs	100	Sufficient for convergence
Batch size	800 (full-batch)	Exact covariance/dCor estimates
Scheduler	Cosine, 5-epoch warmup	Smooth convergence

The full-batch training is computationally trivial since the SDP operates on precomputed 768-dimensional feature vectors, not 3D volumes. Full-batch provides exact (not estimated) covariance and distance correlation statistics.

6.7 Normalisation Scope

Target normalisation statistics (μ, σ) are computed on the training pool (800 subjects) only and applied without recomputation to validation (100), test (150), and the Andalusian cohort. This prevents information leakage and ensures consistent scaling across all downstream uses of the semantic heads.

6.8 Lipschitz Continuity Analysis

With spectral normalisation on both linear layers and GELU activation (Lipschitz constant ≈ 1.13), the global Lipschitz constant of the SDP is bounded by:

$$\mathrm{Lip}(g_\theta) \leq \underbrace{1}_{\mathrm{LN}} \cdot \underbrace{1}_{\mathrm{SN}(\mathbf{W}_1)} \cdot \underbrace{1.13}_{\mathrm{GELU}} \cdot \underbrace{1}_{\mathrm{SN}(\mathbf{W}_2)} \approx 1.13. \tag{32}$$

This ensures that the projection is a near-contraction: $\|\mathbf{z}_1 - \mathbf{z}_2\| \leq 1.13\,\|\mathbf{h}_1 - \mathbf{h}_2\|$. Encoder feature noise is not amplified through the projection, yielding well-conditioned inputs for downstream GP models.

6.9 Quality Targets

Table 14: SDP quality targets (validation set).

Metric	**Minimum**	**Target**	**Blocking?**
Volume R^2	0.80	0.90	Yes
Location R^2	0.85	0.95	Yes
Shape R^2	0.30	0.40	Yes
Max cross-partition correlation	< 0.30	< 0.20	Yes
$\mathrm{dCor}(\mathrm{vol}, \mathrm{loc})$	< 0.20	< 0.10	Yes
Dims with variance > 0.5	$\geq 85\%$	$\geq 95\%$	Yes

Remark 5. *The shape R^2 target of 0.30 was not achieved in any experimental condition ($R^2 \leq 0.11$), motivating the R1 revision to drop the shape partition entirely (Section 9.6).*

7 Phase 3: Cohort Encoding and Harmonisation

7.1 Overview

Phase 3 encodes all MRI volumes from both the BraTS-MEN cross-sectional cohort and the Andalusian longitudinal cohort through the frozen encoder–SDP pipeline, producing per-subject latent vectors and per-patient temporal trajectories suitable for GP-based growth prediction.

The pipeline is fully deterministic (no dropout, model in evaluation mode) and proceeds as:

$$\mathbf{x} \in \mathbb{R}^{4\times 192^3} \xrightarrow{\text{SwinViT (frozen)}} \xrightarrow{\texttt{encoder10}} \xrightarrow{\text{GAP}} \mathbf{h} \in \mathbb{R}^{768} \xrightarrow{g_\theta \text{ (frozen SDP)}} \mathbf{z} \in \mathbb{R}^{128}. \tag{33}$$

7.2 Sliding Window Feature Extraction

For volumes larger than the 128^3 training ROI, or to ensure consistent feature extraction regardless of tumour position, we employ a sliding window approach with tumour-weighted averaging:

$$\mathbf{h} = \frac{\sum_{p=1}^{P} w_p \cdot \mathrm{GAP}\Big(\texttt{encoder10}(\mathbf{x}_p)\Big)}{\sum_{p=1}^{P} w_p}, \quad w_p = \frac{|\mathrm{mask}_p \cap \mathrm{tumour}|}{|\mathrm{patch}_p|}, \tag{34}$$

where $\mathbf{x}_p$ denotes patch p, and w_p weights each patch's contribution by its tumour content fraction. Patches with no tumour ($w_p = 0$) are excluded. This ensures that the feature vector is dominated by tumour-containing regions.

In practice, the feature extraction transforms use the full 192^3 ROI (see Section 3.4), which provides 100% tumour containment for meningiomas without requiring sliding window aggregation. The sliding window protocol is retained for robustness and for use with the Andalusian cohort where acquisition protocols may vary.

7.3 ComBat Harmonisation

Scanner-induced batch effects in multi-site MRI data can introduce spurious variance in the latent space that confounds temporal growth patterns. We assess and optionally apply ComBat harmonisation [57, 58] to the encoded latent vectors.

7.3.1 Assessment Protocol

Before applying ComBat, we assess the degree of scanner-induced shift using:

1. **MMD between cohorts:** $\mathrm{MMD}^2(\mathbf{z}_{\mathrm{BraTS}}, \mathbf{z}_{\mathrm{Andalusian}})$ with permutation test.
2. **UMAP visualisation:** Coloured by site/scanner, to visually assess clustering by acquisition rather than biology.
3. **Kolmogorov–Smirnov tests:** Per-dimension KS tests between sites.

If the MMD permutation test p-value < 0.05, ComBat harmonisation is applied. Otherwise, harmonisation is skipped to avoid unnecessary smoothing of biological signal.

7.3.2 ComBat Formulation

ComBat models each latent dimension j at site s as:

$$z_{ij} = \alpha_j + \mathbf{x}_i^\top \boldsymbol{\beta}_j + \gamma_{sj} + \delta_{sj}\, \epsilon_{ij}, \tag{35}$$

where α_j is the overall mean, $\mathbf{x}_i$ are biological covariates (age, sex if available), γ_{sj} and δ_{sj} are site-specific location and scale effects, and ϵ_{ij} is the residual. Empirical Bayes

estimation of γ and δ provides stable harmonisation even with small site sample sizes. The harmonised value is:

$$z_{ij}^* = \frac{z_{ij} - \hat{\alpha}_j - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j - \hat{\gamma}_{sj}}{\hat{\delta}_{sj}} + \hat{\alpha}_j + \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_j. \tag{36}$$

7.3.3 Scanner Consistency Verification

A critical prerequisite is verifying scanner consistency within each patient's longitudinal series. If a patient was scanned on different scanners across visits, standard ComBat (which assumes one site label per observation) may distort temporal trends. In this case, LongComBat [59], which explicitly models within-subject scanner changes in a mixed-effects framework, should be used instead.

Remark 6. *ComBat's temporal preservation property requires careful verification. For a patient with timepoints t_1, t_2 scanned on the same scanner, the temporal difference $\Delta\mathbf{z} = \mathbf{z}(t_2) - \mathbf{z}(t_1)$ should be preserved up to the site-specific scale factor $\hat{\delta}_s$. This is tested explicitly in the verification protocol (Section 9).*

7.4 Trajectory Construction

For each patient i in the longitudinal cohort, we construct a temporal trajectory by:

1. Encoding all n_i MRI studies through the frozen pipeline, yielding $\{\mathbf{z}_i(t_{ij})\}_{j=1}^{n_i}$.
2. Computing time offsets t_{ij} in months from the patient's first scan: $t_{ij} = (\text{date}_{ij} - \text{date}_{i1})/30.44$.
3. Verifying temporal ordering: $t_{i1} < t_{i2} < \cdots < t_{in_i}$.
4. Extracting the partition subvectors: $\mathbf{z}_{\text{vol}}(t_{ij}) = \mathbf{z}_i(t_{ij})[0:24]$, etc.

The output is a set of per-patient trajectories:

$$\mathcal{T}_i = \Big\{\Big(\mathbf{z}_i(t_{ij}),\ t_{ij}\Big)\Big\}_{j=1}^{n_i}, \quad i = 1, \ldots, N. \tag{37}$$

7.5 Cohort Distribution Comparison

Figure placeholder: Cohort distribution comparison.
UMAP of BraTS-MEN (blue) vs. Andalusian (red) latent vectors.
Left: before ComBat. Right: after ComBat (if applied).

Figure 3: UMAP comparison of BraTS-MEN ($n = 1{,}000$) and Andalusian ($n = 137$) latent representations. The degree of overlap indicates the compatibility of the two cohorts in the learned latent space.

Figure placeholder: Patient trajectories in latent space.
2D UMAP with temporal arrows for 5–10 patients with $n_i \geq 3$.

Figure 4: Latent space trajectories for selected patients with ≥ 3 timepoints. Arrows indicate temporal progression. Smooth, directed trajectories suggest that the latent space captures meaningful temporal dynamics.

8 Growth Prediction: A Three-Stage Complexity Ladder

8.1 Overview and Design Rationale

This section presents the core methodological contribution: a three-stage complexity ladder for meningioma growth prediction, where each stage must justify its existence through improved out-of-sample performance over the preceding one. The key design principle is parsimony: at $N = 31$–58 patients with $\sim$112 total observations, models with more than 2–3 predictor parameters per patient risk overfitting, and complex architectures must demonstrate they capture signal rather than noise.

8.1.1 Why Not a Neural ODE?

The original pipeline design used a Neural ODE [60, 61] for temporal modelling. This was abandoned due to catastrophic overparameterisation: with 33 patients (100 observations, 112 forward pairs), the Neural ODE's $\sim$3,100+ parameters produce a parameter-to-observation ratio of approximately 27.7. The full pivot rationale is documented in [62].

8.1.2 Stage Summary

The three stages, in order of increasing complexity:

Stage 1: Segment tumour $\rightarrow$ extract volume $\rightarrow$ model scalar volume trajectories. Parameters: 2–5 (population level).

Stage 2: Estimate patient-specific latent severity $s_i \in [0, 1] \rightarrow$ model growth as $g(s_i, t; \boldsymbol{\theta})$. Parameters: population $\boldsymbol{\theta}$ + 1 per patient.

Stage 3: BrainSegFounder features $\rightarrow$ PCA $\rightarrow$ GP with ARD kernel. Parameters: population kernel hyperparameters + 3–6 PCA components.

A cross-cutting variance decomposition (Section 9.1) quantifies the marginal ΔR^2 of each stage.

8.2 Stage 1: Segmentation-Based Volumetric Baseline

Stage 1 is the simplest credible growth prediction approach: segment the tumour from each MRI, compute whole-tumour volume in mm^3, and model the resulting scalar volume trajectories. Three models of increasing flexibility are evaluated.

8.2.1 Model 1A: Scalar GP (ScalarGP)

A single GP is fitted to the pooled volume observations across all patients:

$$v_i(t_{ij}) \sim \mathcal{GP}\Big(m(t),\ k_{5/2}(t,t') + \sigma_n^2 \delta(t,t')\Big), \tag{38}$$

where $v_i(t_{ij}) = \log(V_{i,\mathrm{WT}}(t_{ij}) + 1)$ is the log-transformed whole-tumour volume, $m(t) = \beta_0 + \beta_1 t$ is a linear mean function, and $k_{5/2}$ is the Matérn-5/2 kernel (Equation (6)).

Hyperparameters: $\{\beta_0, \beta_1, \sigma_f, \ell, \sigma_n\} = 5$ total, estimated by maximising the log-marginal likelihood.

Covariates: Age and sex can be incorporated into the mean function:

$$m_i(t) = \beta_0 + \beta_1 t + \beta_2 \,\mathrm{age}_i + \beta_3 \,\mathrm{sex}_i, \tag{39}$$

adding 2 parameters. Given the sample size constraints ($N/15 \approx 2$–3 allowed parameters), covariates are included only if they improve LOPO-CV performance.

8.2.2 Model 1B: Linear Mixed-Effects Model (LME)

The LME introduces patient-specific random effects to separate population and individual trends:

$$v_i(t_{ij}) = \underbrace{(\beta_0 + b_{0i})}_{\text{intercept}} + \underbrace{(\beta_1 + b_{1i})}_{\text{slope}} \cdot t_{ij} + \epsilon_{ij}, \tag{40}$$

where (β_0, β_1) are fixed effects (population intercept and slope), $(b_{0i}, b_{1i}) \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega})$ are patient-specific random effects, and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ is residual noise.

Parameter count: 2 fixed effects (β_0, β_1), 3 variance parameters (2 diagonal + 1 off-diagonal of $\mathbf{\Omega}$), 1 residual variance $(\sigma^2) = 6$ total. Estimated via REML using `statsmodels.MixedLM`.

Single-patient inference: The Best Linear Unbiased Predictor (BLUP) [63] provides patient-specific predictions with automatic shrinkage: patients with few observations ($n_i = 2$) receive predictions closer to the population mean, while patients with more observations ($n_i \geq 4$) receive data-driven predictions.

8.2.3 Model 1C: Hierarchical GP (HGP)

The HGP extends the LME by adding nonlinear residuals via a GP:

$$v_i(t) \sim \mathcal{GP}\Big(m(t),\ k_{5/2}(t,t') + \sigma_n^2 \delta(t,t')\Big), \tag{41}$$

where the mean function is the LME population trend:

$$m(t) = \hat{\beta}_0 + \hat{\beta}_1 \cdot t, \tag{42}$$

using fixed effects estimated from the LME. The GP captures nonlinear deviations from the population trend.

Hierarchical hyperparameter sharing: Shared kernel hyperparameters $\boldsymbol{\theta} = (\sigma_f, \ell, \sigma_n)$ are estimated by maximising the pooled log-marginal likelihood across all patients:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{N} \log p\Big(v_i \mid \mathbf{t}_i, \boldsymbol{\theta}\Big). \tag{43}$$

Parameter count: 3 kernel hyperparameters (shared) + 2 mean function parameters (from LME) = 5. Optimised via L-BFGS-B with 5 random restarts.

Model nesting: The GP with linear mean function $\supset$ LME (Section 2.2), ensuring the HGP can never perform worse than the LME on the training data. Improvement must come from the nonlinear Matérn component, which we evaluate via log-marginal likelihood comparison.

8.2.4 Gompertz Mean Function

As an alternative to the linear population mean, we evaluate a Gompertz mean function motivated by the growth biology literature (Section 2.1):

$$m_G(t) = V_0 \exp\left(\frac{\alpha}{\beta}\Big(1 - e^{-\beta t}\Big)\right), \tag{44}$$

where V_0 is the initial volume, α is the initial growth rate, and β is the deceleration parameter. In log-volume space this becomes $\log m_G(t) = \log V_0 + \frac{\alpha}{\beta}(1 - e^{-\beta t})$. The Gompertz mean adds 1 parameter (β) beyond the linear mean, and is evaluated as ablation A2 (Section 9.3).

8.3 Stage 2: Latent Severity Model

Stage 2 posits that inter-patient growth heterogeneity is largely explained by a single latent severity variable $s_i \in [0, 1]$. This is motivated by Vaghi et al.'s [12] finding that a single patient-specific parameter captures most growth variability in tumour populations.

8.3.1 Quantile Transform

To place all patients on a common scale, we apply a quantile transform to both time and growth:

1. **Temporal normalisation.** Within each patient, times are normalised to $[0, 1]$: $\tilde{t}_{ij} = (t_{ij} - t_{i,\min})/(t_{i,\max} - t_{i,\min})$.

2. **Growth normalisation.** Volume changes are mapped to quantiles of the population distribution: $q_{ij} = F_{\text{pop}}(\Delta v_{ij})$, where F_{pop} is the empirical CDF of all observed volume changes across the training set.

The quantile-transformed observations $q_{ij} \in [0, 1]$ represent the relative position of each observation within the population growth distribution.

8.3.2 NLME Formulation

The severity model is a nonlinear mixed-effects model:

$$q_{ij} = g(s_i, \tilde{t}_{ij}; \boldsymbol{\theta}) + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, \sigma^2), \tag{45}$$

where g is a monotonic growth function satisfying boundary conditions $g(s, 0) = 0$ (no growth at baseline) and $g(s, 1) \leq s$ (severity bounds maximum growth). The latent severity $s_i \sim \text{Beta}(\alpha, \beta)$ has a population prior.

8.3.3 Growth Function Options

Three growth functions are evaluated for g:

Sigmoid. The simplest parametric option: $g(s, t; \boldsymbol{\theta}) = s \cdot \sigma(\theta_1(t - \theta_2))$, where σ is the logistic sigmoid and $\boldsymbol{\theta} = (\theta_1, \theta_2)$ controls the steepness and inflection point. Monotonic in s by construction. 2 population parameters.

Reduced Gompertz. Following Vaghi et al. [12]: $g(s, t; \beta) = s \cdot [1 - \exp(-\alpha(s)(1 - e^{-\beta t})/\beta)]$, where $\alpha(s)$ maps severity to growth rate. 1 population parameter (β).

CMNN. A constrained monotonic neural network [23] with non-negative weights ensuring $\partial g/\partial s \geq 0$ everywhere. The most flexible option but with the highest parameter count ($\sim$10–20 weights). Evaluated only if the parametric options prove insufficient.

These are compared in ablation A3 (Section 9.3).

8.3.4 Joint Estimation

The population parameters $\boldsymbol{\theta}$ and patient severities $\{s_i\}_{i=1}^N$ are estimated jointly by maximising the marginal likelihood:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{N} \log \int p(\mathbf{q}_i \mid s_i, \boldsymbol{\theta})\, p(s_i)\, ds_i. \tag{46}$$

For the Beta prior, the inner integral is computed via Gauss–Hermite quadrature or Laplace approximation.

8.3.5 Test-Time Severity Estimation

For a held-out patient with observed trajectory $\{(t_{ij}, q_{ij})\}_{j=1}^{n_i}$, the posterior severity is:

$$p(s_i \mid \mathbf{q}_i, \hat{\boldsymbol{\theta}}) \propto p(\mathbf{q}_i \mid s_i, \hat{\boldsymbol{\theta}})\, p(s_i), \tag{47}$$

which is maximised or integrated over for point and distributional predictions respectively. For patients with $n_i = 2$ (the most common case), the posterior is dominated by the prior, yielding predictions close to the population median—a desirable behaviour analogous to LME shrinkage.

8.3.6 Severity Estimation from Baseline Features

A secondary objective of Stage 2 is to predict severity from baseline (single-timepoint) information, enabling predictions without longitudinal data. Candidate predictors include:

- Clinical covariates: age, sex, tumour location.
- Baseline volume $V_{i,0}$.
- BrainSegFounder features at t_0 (linking to Stage 3).

This is evaluated in ablation A4 (Section 9.3).

8.4 Stage 3: Representation-Learning Augmented Prediction

Stage 3 asks whether high-dimensional deep features from BrainSegFounder capture growth-relevant information beyond what volume and severity already explain. This stage uses the LoRA-adapted encoder (Section 5) and SDP (Section 6) from the original pipeline, but positions them as the final complexity level rather than the primary approach.

8.4.1 Feature Extraction and Compression

The frozen pipeline (encoder + SDP) produces per-scan latent vectors $\mathbf{z} \in \mathbb{R}^{128}$. For growth prediction, these are compressed via within-fold PCA to $\tilde{\mathbf{z}} \in \mathbb{R}^{d_{\text{PCA}}}$ with $d_{\text{PCA}} \in \{3, 4, 5, 6\}$ selected by the sample size constraint and cross-validated performance. PCA is fitted on the training set within each LOPO-CV fold to prevent information leakage.

8.4.2 GP with ARD Kernel

The compressed features are used as inputs to a GP with an ARD kernel:

$$k_{\text{ARD}}(\tilde{\mathbf{z}}, \tilde{\mathbf{z}}'; t, t') = k_{5/2}(t, t') \cdot \prod_{d=1}^{d_{\text{PCA}}} k_{\text{SE}}\left(\frac{\tilde{z}^{(d)} - \tilde{z}'^{(d)}}{\ell_d}\right) + \sigma_n^2 \delta_{ii'} \delta(t, t'), \tag{48}$$

where ℓ_d are per-dimension length-scales. Dimensions with large ℓ_d are effectively deactivated, providing automatic feature selection. The mean function is the Stage 1 population trend.

Parameter count: d_{PCA} ARD length-scales + 3 temporal kernel hyperparameters + mean function parameters. For $d_{\text{PCA}} = 4$: $\sim$9 total hyperparameters, estimated from the pooled marginal likelihood.

8.4.3 Integration with Earlier Stages

Stage 3 does not replace Stages 1 and 2 but augments them. The deep features enter as additional covariates in the GP mean function or as ARD inputs alongside time, while the population trend from Stage 1 remains the prior mean. The marginal contribution is quantified by the variance decomposition (Section 9.1).

8.4.4 SDP with Supervised Residual Targets

An alternative to PCA compression is to use the SDP's supervised partition $\mathbf{z}_{\text{vol}} \in \mathbb{R}^{32}$ directly. The SDP is designed to place volume-relevant information in a structured subspace. However, 32 dimensions is still far too many for GP modelling at $N = 33$. Two options are evaluated:

1. PCA on $\mathbf{z}_{\text{vol}}$ alone (expected to require fewer components than PCA on full $\mathbf{z}$, since the SDP has already concentrated volume-relevant information).

2. Using $\mathbf{z}_{\text{vol}}$ as input to a GP with an ICM coregionalisation matrix (Equation (7)), treating each latent dimension as an output channel. This is the PA-MOGP structure from the original design.

Remark 7 (Stage 3 is conditional)**.** *Stage 3 is attempted only if Stages 1 and 2 leave unexplained variance that could plausibly be captured by additional features. If the variance decomposition shows that volume and severity together explain $R^2 > 0.70$, the marginal room for improvement from deep features is limited, and Stage 3 serves primarily as a negative result confirming parsimony.*

8.5 Leave-One-Patient-Out Cross-Validation

All three stages share a common LOPO-CV protocol. With only 31–58 patients, k-fold CV leaves too few patients per fold for stable model estimation. LOPO-CV uses maximum training data per fold:

Algorithm 1 LOPO-CV Evaluation Protocol (shared across all stages)

```
for i = 1 to N do
    D_train ← {T_j : j ≠ i} {N − 1 patients}
    D_test ← {T_i} {held-out patient}
    Within-fold operations:
        Fit PCA on D_train (Stage 3 only)
        Estimate severity {s_j}_{j≠i} (Stage 2 only)
        Fit model on D_train
        Predict at patient i's observation times
        Record per-patient metrics
end for
Aggregate metrics across N folds
Compute .632+ bootstrap CIs
```

All model fitting, feature transformation (PCA), and severity estimation are performed within each fold. This ensures that LOPO-CV metrics reflect true out-of-sample performance without information leakage.

For each held-out patient i, predictions are evaluated at the patient's observed timepoints. For Stage 2, the held-out patient's severity is re-estimated from their observations using the population model fitted on the remaining patients (Equation (47)).

8.6 Volume Representation and Uncertainty

8.6.1 Volume Representation

All models in Stages 1 and 2 operate on log-transformed whole-tumour volume:

$$v = \log(V_{\mathrm{WT}} + 1), \tag{49}$$

where V_{WT} is the whole-tumour volume in mm^3. This is the single clinically relevant growth endpoint for meningiomas. The log-transformation stabilises variance and makes growth rates approximately additive.

For Stage 3, the SDP semantic head decodes latent predictions to log-volumes:

$$\hat{v} = \pi_{\mathrm{vol}}(\hat{\mathbf{z}_{\mathrm{vol}}}) = \mathbf{W}_v \hat{\mathbf{z}_{\mathrm{vol}}} + \mathbf{b}_v, \quad \hat{v} \in \mathbb{R}^1, \tag{50}$$

with de-normalisation $V_{\mathrm{pred}} = \exp(\hat{v} \cdot \sigma_{\mathrm{vol}} + \mu_{\mathrm{vol}}) - 1$.

8.6.2 Uncertainty Propagation

For GP models that provide predictive variance $\sigma^2_{\hat{v}}$ in log-volume space, the physical volume uncertainty is propagated through the exponential via the delta method:

$$\sigma_{V_{\mathrm{pred}}} \approx V_{\mathrm{pred}} \cdot \sigma_{\hat{v}} \cdot \sigma_{\mathrm{vol}}, \tag{51}$$

where the multiplicative structure reflects the log-normal nature of volume uncertainty. For prediction intervals, we report the 95% credible interval from the GP posterior.

8.7 Failure Recovery Protocol

If GP hyperparameter fitting diverges or produces degenerate results, the following recovery steps are applied in order:

1. Add diagonal jitter to the kernel matrix: $\mathbf{K} \leftarrow \mathbf{K} + 10^{-6}\,\mathbf{I}$.
2. Constrain length-scale bounds: $\ell \in [1, 120]$ months.
3. Increase L-BFGS-B random restarts from 5 to 10.
4. Fall back to a simpler kernel (Matérn-5/2 $\rightarrow$ SE).
5. If all fail: report per-dimension diagnostics and halt.

For Stage 2 severity estimation, if the joint optimisation fails to converge:

1. Reduce the growth function complexity (CMNN $\rightarrow$ sigmoid).
2. Initialise severities from the empirical growth rate ranking.
3. Fix population parameters and estimate severities only (profile likelihood).

9 Evaluation: Variance Decomposition and Quality Assessment

9.1 Variance Decomposition Framework

The central analytical contribution of this thesis is a variance decomposition quantifying the marginal predictive value of each complexity stage. We define a nested model hierarchy:

Table 15: Nested model hierarchy for variance decomposition.

Model	Description	Params	Inputs
M_0	Population mean only	2	t
M_1	ScalarGP (Stage 1)	5	t
M_2	LME (Stage 1)	6	t, patient ID
M_3	HGP (Stage 1)	5	t, patient ID, LME mean
M_4	Severity model (Stage 2)	$\sim 3 + N$	t, s_i
M_5	Deep features (Stage 3)	~ 9	t, $\tilde{\mathbf{z}}$

9.1.1 Marginal ΔR^2

For each successive model, the marginal contribution is:

$$\Delta R_k^2 = R^2(M_k) - R^2(M_{k-1}), \tag{52}$$

where R^2 is computed from LOPO-CV predictions (out-of-sample). The ΔR^2 values are not necessarily positive: a more complex model may perform worse out-of-sample due to overfitting. A negative ΔR^2 is strong evidence that the additional complexity is not warranted at the current sample size.

9.1.2 Statistical Significance

To determine whether a ΔR^2 is statistically significant, we use two complementary tests:

Paired permutation test. For each LOPO-CV fold i, compute the per-fold squared error for models M_k and M_{k-1}. Under the null hypothesis $H_0 : \Delta R^2 = 0$, the fold-level error differences are exchangeable. We permute the signs of $\{\mathrm{SE}_i^{(k)} - \mathrm{SE}_i^{(k-1)}\}_{i=1}^N$ over $B = 10{,}000$ iterations and compute the p-value as the fraction of permutations with $|\Delta\overline{\mathrm{SE}}| \geq |\Delta\overline{\mathrm{SE}}_{\mathrm{obs}}|$.

BCa bootstrap CIs. Bias-corrected and accelerated (BCa) bootstrap confidence intervals [64] for R^2 and ΔR^2, using $B = 2{,}000$ resamples of the LOPO-CV fold results. The 95% CI for ΔR^2 that excludes zero provides an additional significance criterion.

9.1.3 Decision Rule

The complexity ladder advances to the next stage only if:

1. $\Delta R^2 > 0$ (positive marginal contribution).
2. The paired permutation test yields $p < 0.05$.
3. The 95% BCa bootstrap CI for ΔR^2 excludes zero.

If any condition fails, the simpler model is preferred and the higher stage is reported as a negative result.

9.2 Quality Targets per Stage

Each stage has specific quality targets that must be met before advancing.

9.2.1 Stage 1: Volumetric Growth Models

Table 16: Stage 1 quality targets.

Metric	Target	Minimum	Status
ScalarGP LOPO completes (33 folds)	All	All	*[to fill]*
LME LOPO R^2	> 0.30	> 0	*[to fill]*
HGP LOPO $R^2 \geq$ ScalarGP R^2	Yes	—	*[to fill]*
GP calibration (95% CI coverage)	0.90–0.98	≥ 0.80	*[to fill]*

9.2.2 Stage 2: Latent Severity Model

Table 17: Stage 2 quality targets.

Metric	Target	Minimum	Status
Growth function monotonicity	Verified	Verified	*[to fill]*
Boundary condition $g(s, 0) = 0$	Exact	Exact	*[to fill]*
Severity spread ($\text{std}(s_i) > 0.1$)	> 0.15	> 0.10	*[to fill]*
LOPO $R^2 \geq$ Stage 1 best	Yes	—	*[to fill]*
ΔR^2 significant ($p < 0.05$)	Yes	—	*[to fill]*

9.2.3 Stage 3: Representation Learning

Table 18: Stage 3 quality targets.

Metric	Target	Minimum	Status
SDP Vol R^2 on validation	≥ 0.90	≥ 0.80	*[to fill]*
PCA components retained	3–6	≥ 3	*[to fill]*
LOPO $R^2 \geq$ Stage 2	Yes	—	*[to fill]*
ΔR^2 significant ($p < 0.05$)	Yes	—	*[to fill]*
ARD length-scales interpretable	Yes	—	*[to fill]*

9.2.4 Cross-Stage Targets

Table 19: Cross-stage quality targets.

Metric	Target	Minimum	Status
Best LOPO Vol R^2 (any model)	≥ 0.50	≥ 0.30	*[to fill]*
Calibration (95% CI, best GP)	0.90–0.98	≥ 0.80	*[to fill]*
Per-patient r ($n_i \geq 3$)	≥ 0.80	≥ 0.60	*[to fill]*
Variance decomposition complete	All stages	Stage 1	*[to fill]*

9.2.5 Phase 1 (LoRA) and Phase 2 (SDP) Quality Targets

The LoRA and SDP phases (relevant to Stage 3) retain their original quality targets for internal validation:

Table 20: Phase 1 quality targets (encoder adaptation).

Metric	Target	Minimum	Status
Dice (WT) on `lora_val`	≥ 0.85	≥ 0.80	*[to fill]*
Dice improvement over frozen BSF	≥ 0.05	≥ 0.02	*[to fill]*
Linear probe Vol R^2 (adapted)	≥ 0.50	≥ 0.30	*[to fill]*

9.3 Ablation Study Design

Six ablation experiments evaluate key design decisions across the three stages:

Table 21: Ablation matrix (A1–A6).

ID	Variable	Conditions	Primary metric
A1	Segmentation model	{frozen BSF, LoRA-adapted, fine-tuned}	Stage 1 Vol R^2 (LOPO
A2	GP mean function	{zero, linear, Gompertz}	Stage 1 Vol R^2 (LOPO
A3	Growth function (Stage 2)	{sigmoid, reduced Gompertz, CMNN}	Stage 2 R^2 (LOPO)
A4	Severity estimation inputs	{clinical, clinical+deep, deep-only}	Severity–growth correla
A5	LoRA rank (Stage 3)	$\{2, 4, 8, 16, 32\}$	Dice, probe R^2
A6	PCA components (Stage 3)	$\{3, 4, 5, 6, 8\}$	Stage 3 R^2 (LOPO)

9.3.1 A1: Segmentation Model Comparison

Stage 1 requires accurate tumour segmentation to extract volume. This ablation compares three segmentation sources:

- **Frozen BSF:** Direct application of the pretrained BrainSegFounder checkpoint without any meningioma-specific adaptation.
- **LoRA-adapted BSF:** After Phase 1 LoRA training on BraTS-MEN 2024 (Section 5).
- **Fine-tuned BSF:** Full fine-tuning of the decoder (encoder frozen), as a more expensive but potentially more accurate alternative.

The segmentation quality directly affects Stage 1 volume accuracy and therefore all downstream predictions. This ablation determines whether LoRA adaptation is necessary for growth prediction (as opposed to segmentation quality *per se*).

9.3.2 A2: Gompertz versus Linear Mean

Motivated by the growth biology literature (Section 2.1), this ablation tests whether a Gompertz mean function (Equation (44)) improves the HGP over a linear mean. The Gompertz mean adds one parameter (β, the deceleration constant) and encodes the prior that meningioma growth decelerates—a strong biological prior that may be especially valuable at small N.

9.3.3 A6: PCA Components for Stage 3

The number of PCA components d_{PCA} controls the complexity–information tradeoff in Stage 3. Too few components discard growth-relevant information; too many introduce noise dimensions that the GP cannot reliably fit at $N = 33$.

9.4 Model Comparison Results

Table 22: Stage 1 model comparison (LOPO-CV, N folds).

Model	**Params**	**Vol R^2**	**Vol MAE**	**Calibration**	**Per-patient r**
ScalarGP	5	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
LME	6	*[to fill]*	*[to fill]*	—	*[to fill]*
HGP	5	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*

Table 23: Cross-stage model comparison (LOPO-CV, N folds).

Stage	**Best model**	**Vol R^2**	**ΔR^2**	**p-value**	**95% CI**
Stage 1	*[to fill]*	*[to fill]*	—	—	*[to fill]*
Stage 2	Severity	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*
Stage 3	Deep GP	*[to fill]*	*[to fill]*	*[to fill]*	*[to fill]*

9.5 Publication Figures

Figure placeholder: Variance decomposition bar chart.
Stacked ΔR^2 contributions from Stage 1, Stage 2, and Stage 3,
with BCa bootstrap 95% CIs on each increment.

Figure 5: Variance decomposition across the three-stage complexity ladder. Each bar segment shows the marginal ΔR^2 contributed by each stage. Error bars are BCa bootstrap 95% CIs. A segment crossing zero indicates that the stage does not improve out-of-sample prediction.

Figure placeholder: Volume prediction scatter.
Predicted vs. actual ΔV for all LOPO-CV test pairs,
coloured by model/stage.

Figure 6: Predicted vs. actual volume change (ΔV, mm^3) across all LOPO-CV held-out observation pairs, coloured by stage. The diagonal represents perfect prediction.

Figure placeholder: Trajectory prediction with uncertainty.
3–4 example patients showing predicted trajectory
(mean ± 95% CI) overlaid on actual observations.

Figure 7: Growth trajectory predictions for selected patients from the best-performing model. Solid lines show posterior mean; shaded regions show 95% credible intervals. Dots are observed volumes. Uncertainty naturally widens with extrapolation distance.

Figure placeholder: Severity distribution.
Histogram of estimated s_i values across patients,
coloured by growth subgroup (stable, slow, fast).

Figure 8: Distribution of estimated latent severity s_i across the longitudinal cohort. Patients stratify into the three growth subgroups identified by Engelhardt et al. [6]: stable ($s < 0.3$), slow growth ($0.3 \leq s \leq 0.7$), and fast growth ($s > 0.7$). The Beta prior is overlaid in grey.

Figure placeholder: Model comparison bar chart.
R^2, MAE, and calibration for each model across all stages.

Figure 9: Head-to-head model comparison across three metrics, with models grouped by stage. Error bars represent bootstrap 95% confidence intervals computed over the LOPO-CV folds.

9.6 R1 Methodology Revision

Empirical findings from the initial pipeline (before the complexity ladder reframing) motivated three principled revisions.

9.6.1 Shape Partition Failure

Across all experimental conditions, the shape partition achieved $R^2 \leq 0.11$ for linear probes and approximately zero for RBF probes. Cross-domain shape transfer was uniformly catastrophic ($R^2 = -2.00$). This failure is biologically grounded: gliomas are infiltrative and irregular, while meningiomas are well-circumscribed and dural-based.

Table 24: Shape probe R^2 across experimental conditions.

Condition	**Linear R^2**	**RBF R^2**	**Cross-domain**
Frozen baseline	0.08	≈ 0	-2.00
MEN LoRA r=8	0.11	≈ 0	-2.00
Dual LoRA r=8	-0.06	≈ 0.15	-2.00

Decision: Drop the shape partition from the SDP (Stage 3).

9.6.2 Location: Temporally Static

Meningiomas are extra-axial, dural-attached tumours that do not migrate. Modelling location as a time-varying latent dimension is statistically vacuous. Within-domain location R^2 is reasonable (0.35 MEN, 0.68 GLI), confirming the information is present in the features but does not need temporal modelling.

Decision: Relocate location from temporal latent dimension to static GP covariate.

9.6.3 Volume Target Simplification

The 4-dimensional volume target $[\log V_{\text{total}}, \log V_{\text{NCR}}, \log V_{\text{ED}}, \log V_{\text{ET}}]$ is problematic because sub-compartment segmentation quality is poor (TC Dice 0.40 on MEN) and whole-tumour volume is the only clinically relevant growth endpoint for meningiomas.

Decision: Reduce to a single target $v^* = \log(V_{\text{WT}} + 1) \in \mathbb{R}^1$.

9.6.4 The Complexity Ladder as a Methodological Response

The R1 revision also motivated the fundamental reframing from a linear pipeline (LoRA $\rightarrow$ SDP $\rightarrow$ GP on latent trajectories) to the three-stage complexity ladder. The key insight was that the marginal benefit of LoRA adaptation for downstream feature quality was $\Delta R^2 = +0.016$ (Section 9.7), raising the question of whether the entire representation learning pipeline was necessary. The complexity ladder addresses this question directly by placing simple volumetric models first and requiring each additional stage to demonstrate measurable improvement.

9.7 LoRA Marginality

A notable finding from the initial pipeline is the marginal benefit of LoRA adaptation for downstream feature quality. The frozen BrainSegFounder baseline achieves $R^2 = 0.794$ for volume probing, while the best LoRA condition adds only $\Delta R^2 = +0.016$ (within noise). This suggests that BrainSegFounder's SSL pretraining on 41,400+ subjects already produces features with strong meningioma volume decodability.

Several hypotheses explain this observation:

1. **SSL pretraining captures domain-general features.** The UK Biobank pretraining stage exposes the encoder to massive anatomical diversity, producing features that generalise across brain pathologies.

2. **LoRA's low-rank constraint limits adaptation.** With $r = 8$ and $< 0.32\%$ trainable parameters, LoRA may not have sufficient capacity to substantially restructure the 768-dimensional feature space.

3. **Segmentation is a weak proxy for semantic quality.** Improving segmentation Dice does not necessarily improve the linear decodability of aggregate semantic properties.

This finding motivates the complexity ladder design: if LoRA provides marginal benefit for feature quality, the entire Stage 3 representation learning pipeline may provide

marginal benefit for growth prediction. The variance decomposition will quantify this directly.

10 Conclusion

10.1 Summary of Contributions

This thesis has presented a three-stage complexity ladder for meningioma growth forecasting from longitudinal multi-modal 3D MRI. Rather than proposing a single complex pipeline and reporting its metrics, we systematically evaluate whether each successive level of model sophistication provides measurable out-of-sample improvement at $N = 31$–58 patients. The key contributions and findings are:

1. **The first out-of-sample meningioma growth prediction benchmark.** Using LOPO-CV on a longitudinal cohort, we provide the first reported R^2, MAE, and calibration metrics for meningioma volume prediction. The volumetric baseline (Stage 1: ScalarGP, LME, HGP) establishes a reference point that, to our knowledge, did not previously exist in the literature. The HGP with population linear mean from the LME provides calibrated uncertainty with automatic shrinkage toward the population trend for patients with sparse observations.

2. **A latent severity framework for growth characterisation.** Stage 2 formulates meningioma growth heterogeneity as a latent variable problem connected to Item Response Theory and the reduced Gompertz model [12]. A single patient-specific severity parameter $s_i \in [0, 1]$ captures inter-patient variability while respecting the sample size constraint of 2–3 parameters per predictor. The framework provides interpretable patient stratification aligned with the three growth subgroups identified by Engelhardt et al. [6].

3. **Systematic assessment of representation learning value.** By placing the full BrainSegFounder + LoRA + SDP pipeline as Stage 3, we directly quantify whether 768-dimensional deep features (compressed via PCA to 3–6 dimensions) capture growth-relevant information beyond what volume and severity already explain. The marginal benefit of LoRA adaptation for feature quality ($\Delta R^2 = +0.016$ on semantic probes) motivates scepticism that the full representation learning pipeline is necessary for this task.

4. **Variance decomposition as an analytical framework.** The cross-cutting analysis decomposes LOPO-CV prediction variance across the three stages using paired permutation tests and BCa bootstrap CIs. This provides not only a model selection

criterion but also a methodological template for evaluating complexity–benefit trade-offs in small-sample longitudinal prediction problems more broadly.

5. **Empirically-driven methodology revision.** Rigorous evaluation revealed that shape disentanglement fails ($R^2 \leq 0.11$), meningioma location is temporally static, and sub-compartment volume targets are label-dependent and clinically irrelevant. These findings motivated a principled simplification and the reframing from a linear pipeline to the complexity ladder.

10.2 Limitations

10.2.1 Cohort Size

The longitudinal cohort ($N = 31$–58 patients, $\sim$112 observations) is the primary bottleneck. With approximately 57.6% of patients having only 2 studies, per-patient growth trajectories are sparse. The GP and LME frameworks handle this via automatic shrinkage toward the population mean, but predictions for 2-observation patients are necessarily dominated by population-level dynamics rather than individual-specific growth rates. The sample size also limits the statistical power of the variance decomposition: Varoquaux [26] showed that CV error estimates have standard deviations of $\sim\pm10\%$ at $N = 100$, and uncertainty is larger still at $N = 33$.

10.2.2 Ordinal Time

At the time of writing, exact acquisition timestamps for the longitudinal cohort are pending. If only ordinal timepoint indices are available (scan 1, scan 2, scan 3, ...) rather than calendar dates, the temporal models operate on a surrogate time axis. This does not invalidate the complexity ladder comparison (all models use the same time representation) but limits the clinical interpretability of growth rate estimates and length-scale hyperparameters.

10.2.3 LoRA Marginality and Stage 3 Viability

The marginal benefit of LoRA adaptation ($\Delta R^2 = +0.016$ for semantic probing) suggests that the entire Stage 3 pipeline may provide limited value at the current sample size. This is, however, a finding rather than a failure: establishing that simple volumetric models suffice is a valuable negative result that can save computational resources in clinical deployment.

10.2.4 Single-Site Longitudinal Data

The longitudinal cohort is collected from hospitals in a single region (Andalusia, Spain), which may limit generalisability to other populations, scanner manufacturers, and ac-

quisition protocols. ComBat harmonisation addresses scanner batch effects but cannot correct for biological population differences.

10.2.5 Absence of Clinical Validation

The pipeline has not been validated against clinical outcomes (intervention decisions, actual tumour behaviour at longer follow-up). Volume prediction R^2 and calibration metrics are necessary but not sufficient for clinical utility. A positive growth prediction does not, by itself, determine the appropriate clinical action.

10.3 Future Work

10.3.1 Cohort Expansion to $N \geq 50$

The most impactful improvement would be expanding the longitudinal cohort. Multi-centre collaborations targeting $N \geq 50$ patients would: (i) tighten the variance decomposition CIs, enabling confident ΔR^2 conclusions; (ii) potentially unlock value in Stage 2 and 3 by providing sufficient data for more complex models; (iii) enable subgroup analyses by tumour grade, location, and patient demographics.

10.3.2 Exact Timestamps

Obtaining exact acquisition dates would enable physically meaningful growth rate estimation, improve GP length-scale interpretability, and allow direct comparison with the volumetric growth rates reported in the clinical literature [3, 11, 6].

10.3.3 Clinical Decision Support Integration

The GP posterior's calibrated uncertainty is directly suitable for clinical decision support: the probability that a tumour will exceed a volume threshold V_{critical} within a specified time horizon can be computed analytically from the Gaussian predictive distribution:

$$P(V(t_{\text{future}}) > V_{\text{critical}}) = 1 - \Phi\left(\frac{\log(V_{\text{critical}} + 1) - \mu_*(t_{\text{future}})}{\sigma_*(t_{\text{future}})}\right), \tag{53}$$

where Φ is the standard normal CDF and (μ_*, σ_*) are the GP posterior mean and standard deviation in log-volume space. Integrating this into a clinical workflow would require prospective validation and regulatory approval.

10.3.4 Severity-Guided Surveillance

If the severity model (Stage 2) provides robust patient stratification, it could inform surveillance scheduling: high-severity patients receive shorter follow-up intervals, while

low-severity patients can be monitored less frequently. This requires validation of the severity–growth relationship at longer follow-up horizons.

10.3.5 Foundation Model Scaling

BrainSegFounder is a relatively small foundation model (BSF-Tiny, 62M parameters). Larger variants or more recent foundation models pretrained on broader datasets may provide stronger initial features, potentially shifting the Stage 3 ΔR^2 from marginal to significant. However, this hypothesis must be tested empirically rather than assumed.

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